

# CICADA AGI

## Strategic Development Roadmap

Building the Future of Autonomous Security Research

Security Research Laboratory  
Zero-Day Discovery and Exploit Development

2026 - 2029

# Executive Summary

Cicada AGI represents a transformative initiative to establish a world-class security research laboratory focused on zero-day vulnerability discovery and exploit development. This strategic roadmap outlines a phased approach to building autonomous security research capabilities, progressing from our current red teaming application foundation to a fully automated, AI-driven discovery pipeline that operates continuously to identify and develop proof-of-concept exploits for both novel vulnerabilities and publicly disclosed CVEs.

Our vision centers on developing specialized AI agents capable of autonomous reconnaissance, reasoning, and exploitation. By leveraging local large language models for cost-effectiveness while maintaining the flexibility to incorporate premium models when necessary, we aim to build a sustainable competitive advantage. The roadmap progresses through five distinct phases, each building upon the previous to create increasingly sophisticated autonomous capabilities, ultimately reaching a "Manager of Managers" model where human researchers orchestrate swarms of specialized AI agents.

Throughout this journey, we maintain a commitment to preserving human expertise for out-of-the-box research and novel approaches that require creative reasoning beyond current AI capabilities. This balanced approach ensures we capture the efficiency gains of automation while retaining the innovative capacity that drives breakthrough discoveries.

## Vision Statement

Our laboratory is dedicated to discovering zero-day vulnerabilities through rigorous research and innovative methodologies. We aspire to become one of the premier security research institutions globally, recognized for our contributions to vulnerability disclosure and our commitment to advancing the field of autonomous security testing. Our approach combines cutting-edge artificial intelligence with deep security expertise to create a sustainable pipeline of vulnerability research that benefits our clients and the broader security community.

The core philosophy driving Cicada AGI centers on building proprietary capabilities rather than relying on public tools. We are developing our own Agentic Hacking Framework that chains multiple specialized models, each optimized for specific tasks such as code review, network traffic analysis, and exploit writing. This modular architecture enables continuous improvement and adaptation as new techniques emerge in the security landscape.

## Strategic Pillars

Autonomous Research Pipeline: Development of a path-guided iterative pipeline featuring autonomous reconnaissance, mapping, reasoning agents, and exploit development capabilities for proof-of-concept creation.

CVE-Driven Development: Systematic analysis of reported CVEs to build proof-of-concept exploits, providing early tactical advantages for client protection when public PoCs are unavailable.

Cost-Effective AI Operations: Strategic deployment of local LLMs to minimize operational costs while maintaining flexibility to integrate premium models for critical tasks requiring superior performance.

Proprietary Framework Development: Construction of a custom Agentic Hacking Framework that orchestrates specialized models, reducing dependency on public tools and enabling tailored optimization.

Continuous Operation: Establishment of 24/7 automated discovery pipelines powered by in-house trained models, with human researchers focused on model tuning and edge-case research.

Human-AI Collaboration: Preservation of dedicated talent for innovative, out-of-the-box research that requires creative thinking beyond current AI reasoning capabilities.

## Phase 1: Foundation and MVP Development

Timeline: Q1 2026 - Q2 2026 (Current Phase)

The foundation phase centers on establishing our core technology infrastructure through the Cicada AGI Red Teaming Tool. This application serves as the primary platform for integrating third-party LLMs with penetration testing workflows, providing a robust foundation for subsequent development phases. The current implementation leverages existing AI capabilities while our team develops the expertise and infrastructure necessary for in-house model development.

### Current Platform Capabilities

The Cicada AGI Red Teaming Tool represents a comprehensive security testing platform designed for enterprise-grade operations. Built on a microservices architecture supporting horizontal scaling, the platform provides complete control over deployment and data through self-hosted infrastructure. All operations execute within sandboxed Docker environments ensuring complete isolation, while the AI-powered agent automatically determines and executes penetration testing steps with optional execution monitoring and intelligent task planning for enhanced reliability.

| Category                | Capabilities                                                                                                  |
|-------------------------|---------------------------------------------------------------------------------------------------------------|
| Security Infrastructure | Docker sandboxed operations, complete isolation, secure bearer token authentication for REST and GraphQL APIs |

|                       |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| Autonomous Operations | AI-powered agent with automatic step determination, execution monitoring, intelligent task planning |
| Tool Suite            | 20+ professional security tools including nmap, metasploit, sqlmap, and specialized utilities       |
| Intelligence Systems  | Smart memory system, Graphiti-powered Neo4j knowledge graph, semantic relationship tracking         |
| External Integrations | Tavily, Traversaal, Perplexity, DuckDuckGo, Google Custom Search, Sploitius, Searxng                |
| Monitoring            | Comprehensive logging, Grafana/Prometheus integration, real-time system observation                 |
| Data Persistence      | PostgreSQL with pgvector extension, persistent command and output storage                           |

Table 1: Cicada AGI Red Teaming Tool Core Capabilities

### LLM Provider Support

The platform supports flexible authentication across more than ten LLM providers, enabling strategic selection based on task requirements and cost considerations. Supported providers include OpenAI, Anthropic, Google AI/Gemini, AWS Bedrock, Ollama for local deployments, DeepSeek, GLM, Kimi, Qwen, and custom endpoints. Aggregator services such as OpenRouter and DeepInfra provide additional flexibility for model selection. For production local deployments, the platform includes comprehensive guidance for vLLM deployment with Qwen3.5-27B-FP8 configurations, enabling cost-effective high-performance inference.

### Phase 1 Objectives

- Complete MVP development and transition to production environment
- Establish baseline performance metrics for third-party LLM integration
- Document operational workflows and identify optimization opportunities
- Build team expertise in AI-driven security testing methodologies
- Create feedback loops for continuous platform improvement

## Phase 2: In-House Benchmark Development

Timeline: Q2 2026 - Q3 2026

The benchmarking phase establishes proprietary evaluation frameworks specifically designed for security research use cases. While general-purpose benchmarks exist for LLM evaluation, the unique requirements of security testing demand specialized metrics that capture performance across reconnaissance, vulnerability identification, exploit development, and reporting tasks. This phase creates the evaluation infrastructure

necessary for informed model selection and subsequent distillation efforts.

## Benchmark Development Strategy

Our benchmark development approach focuses on creating task-specific evaluation suites that measure practical security research capabilities. These benchmarks will assess model performance across multiple dimensions including code review accuracy, network traffic interpretation, vulnerability pattern recognition, exploit logic construction, and technical documentation generation. Each benchmark category includes both synthetic test cases and anonymized real-world scenarios from previous security research engagements.

The evaluation framework incorporates automated scoring mechanisms where feasible, supplemented by expert review for complex reasoning tasks. Performance metrics capture both accuracy and efficiency, enabling cost-benefit analysis across model options. Results feed directly into the model selection process for Phase 3 distillation efforts, ensuring our in-house models target the highest-impact capabilities.

## Key Benchmark Categories

**Reconnaissance Analysis:** Evaluation of information gathering and target profiling capabilities across network, web application, and infrastructure targets.

**Vulnerability Identification:** Assessment of pattern recognition for common and novel vulnerability classes, including misconfiguration detection and logic flaw identification.

**Exploit Development:** Measurement of code generation quality for proof-of-concept exploit development across multiple vulnerability types.

**Traffic Analysis:** Evaluation of network traffic interpretation for identifying security-relevant patterns and anomalous behaviors.

**Report Generation:** Assessment of technical documentation quality, including vulnerability descriptions, reproduction steps, and remediation recommendations.

## Phase 2 Deliverables

Comprehensive benchmark suite covering all major security research tasks

Automated evaluation pipeline with scoring and reporting capabilities

Comparative analysis of available third-party LLMs on security-specific tasks

Cost-performance matrix for model selection decisions

Recommendations for distillation target models and capabilities

# Phase 3: In-House Model Development and Distillation

Timeline: Q3 2026 - Q2 2027

The model development phase represents a critical transition from reliance on third-party LLMs to proprietary in-house capabilities. Using knowledge distillation techniques, we will transfer specialized security research capabilities from the highest-performing commercial models to locally-deployed models optimized for our specific use cases. This approach enables significant cost reduction while maintaining or improving task-specific performance through targeted training.

### Distillation Methodology

Knowledge distillation leverages the superior performance of large commercial models to train smaller, more efficient models on specialized tasks. Our approach begins with comprehensive data collection using the benchmark evaluation phase to identify optimal teacher models for each capability category. The distillation process generates high-quality training data through teacher model inference on security research tasks, capturing both outputs and reasoning patterns.

Training proceeds through iterative refinement, with each generation evaluated against our benchmark suite to ensure capability preservation. The architecture selection balances model size against deployment constraints, targeting configurations that enable efficient local inference while maintaining sufficient capacity for complex reasoning tasks. Initial efforts focus on single-task models optimized for specific capabilities, with subsequent integration into unified multi-task architectures.

| Stage            | Activities                                                    | Duration   |
|------------------|---------------------------------------------------------------|------------|
| Data Collection  | Generate training data using teacher models on security tasks | 4-6 weeks  |
| Initial Training | Train specialized models for individual capability categories | 6-8 weeks  |
| Integration      | Combine specialized models into unified architecture          | 4-6 weeks  |
| Optimization     | Fine-tune for performance and efficiency improvements         | 8-12 weeks |
| Validation       | Comprehensive benchmark evaluation and deployment preparation | 4-6 weeks  |

Table 2: Model Development Stages and Timeline

### Infrastructure Requirements

Model development requires significant computational infrastructure for both training and inference workloads. The infrastructure strategy balances initial investment against operational costs, with options including on-premises GPU clusters, cloud-based training instances, and hybrid configurations. Inference infrastructure scales with operational demands, with initial deployment targeting single-node configurations suitable for research team usage, expanding to multi-node clusters as production workloads increase.

## Phase 4: Autonomous Discovery Pipeline

Timeline: Q2 2027 - Q4 2027

The autonomous discovery pipeline phase establishes continuous, automated security research operations powered by our in-house trained models. This represents the realization of our core vision: a 24/7 discovery engine that autonomously conducts reconnaissance, identifies vulnerabilities, and develops proof-of-concept exploits with minimal human intervention. Human researchers transition to oversight, tuning, and edge-case research roles.

### Pipeline Architecture

The autonomous pipeline implements a path-guided iterative architecture where specialized agents collaborate on complex security research tasks. The architecture comprises distinct agent types including reconnaissance agents for target mapping, analysis agents for vulnerability identification, reasoning agents for exploitation strategy development, and exploit agents for proof-of-concept creation. Each agent type leverages models optimized for its specific task category, with orchestration layers managing inter-agent communication and workflow progression.

Iterative refinement loops enable the pipeline to handle complex targets requiring multiple exploration paths. When initial approaches fail, reasoning agents analyze outcomes and propose alternative strategies, enabling adaptive behavior without human intervention. Memory systems capture successful approaches for future reference, building organizational knowledge that improves pipeline performance over time. The knowledge graph integration enables semantic relationship tracking across research sessions, identifying patterns that inform future discovery efforts.

### CVE Processing Workflow

A dedicated pipeline component focuses on newly disclosed CVEs, rapidly analyzing vulnerability details and attempting proof-of-concept development. This capability provides significant tactical value when public PoCs are unavailable, enabling our clients to assess exposure and implement mitigations before exploitation occurs in the wild. The workflow begins with automated CVE monitoring and classification, followed by technical analysis to understand vulnerability mechanics and identify potential exploitation approaches.

Exploit development proceeds through iterative refinement, with the pipeline attempting multiple approaches based on vulnerability characteristics. Successful exploits undergo validation and documentation before delivery to clients. The entire process operates on compressed timelines, targeting PoC availability within hours of CVE disclosure for high-priority vulnerabilities. This capability distinguishes our laboratory from

competitors reliant on public exploit availability.

## Phase 4 Milestones

Deployment of continuous discovery pipeline with autonomous operation capabilities

Integration of in-house models across all pipeline components

Establishment of CVE monitoring and rapid PoC development workflow

Implementation of comprehensive logging and performance tracking systems

Achievement of target discovery metrics with minimal human intervention

## Phase 5: Manager of Managers Model

Timeline: Q1 2028 - Q4 2029

The Manager of Managers model represents the culmination of our autonomous research vision, establishing an organizational structure where senior human researchers orchestrate teams of AI agents. This model maximizes research throughput by enabling a small number of expert researchers to manage multiple parallel research streams, with AI agents handling execution while humans provide strategic direction, quality oversight, and creative problem-solving for edge cases.

### Organizational Architecture

The Manager of Managers architecture creates a hierarchical structure where human researchers serve as strategic directors, managing swarm teams of specialized AI agents. Each swarm team focuses on specific research domains or target types, with agents collaborating to accomplish objectives defined by their human manager. The architecture enables parallel execution across multiple research streams while maintaining coherent strategic direction.

Swarm teams comprise multiple agent types with complementary capabilities. Research swarms focus on vulnerability discovery and analysis, development swarms handle exploit creation and tooling, and infrastructure swarms manage operational support functions. Human managers allocate resources, set priorities, review results, and intervene when agents encounter challenges beyond their capabilities. This structure scales research capacity without proportional increases in expert personnel.

### Human Expertise Preservation

A core principle throughout our roadmap is the preservation of human expertise for out-of-the-box research and creative problem-solving. While AI excels at pattern recognition and systematic execution, certain research challenges require creative thinking and novel approaches that current AI capabilities cannot replicate. Our



organizational model reserves capacity specifically for human-driven innovation, ensuring we maintain competitive advantage in areas requiring genuine creativity.

Human researchers in the Manager of Managers model focus on several high-value activities. Strategic planning involves identifying promising research directions and allocating resources across initiatives. Quality assurance ensures AI-generated outputs meet professional standards before client delivery. Innovation research explores novel techniques and approaches that may later be integrated into automated pipelines. Edge-case handling addresses unique challenges that fall outside AI training distributions. This balanced approach captures efficiency gains while preserving innovative capacity.

## Agentic Hacking Framework Architecture

Central to our technical strategy is the development of a proprietary Agentic Hacking Framework that orchestrates multiple specialized models. Unlike reliance on public tools, this framework enables tailored optimization for our specific research requirements while maintaining flexibility for continuous improvement as new techniques emerge. The framework architecture supports modular model integration, enabling upgrades and replacements without disrupting operational workflows.

### Specialized Model Categories

**Code Review Model:** Optimized for static analysis, vulnerability identification in source code, and security pattern recognition. Capable of analyzing codebases across multiple programming languages and frameworks.

**Network Traffic Model:** Specialized for interpreting network communications, identifying security-relevant patterns, and detecting anomalous behaviors in traffic captures.

**Exploit Writing Model:** Focused on proof-of-concept development, shellcode generation, and exploit chain construction. Trained on vulnerability patterns and exploitation techniques.

**Reconnaissance Model:** Optimized for information gathering, target profiling, and attack surface analysis. Capable of synthesizing data from multiple sources for comprehensive target understanding.

**Reasoning Model:** Handles complex decision-making, strategy development, and multi-step planning. Enables adaptive behavior when encountering unexpected scenarios.

### Framework Capabilities

The framework provides several key capabilities essential for autonomous security research. Model chaining enables sequential execution where outputs from one model inform inputs to another, creating sophisticated analysis pipelines. Parallel execution supports simultaneous operation of independent tasks, maximizing throughput on complex targets. State management captures intermediate results and enables resumption after

interruptions. Feedback loops allow models to refine outputs based on validation results, improving output quality through iteration.

Integration interfaces connect the framework with external tools and data sources, enabling comprehensive research capabilities. The tool integration layer provides standardized interfaces for security tools including network scanners, web application testers, and exploitation frameworks. Data source connectors enable access to vulnerability databases, threat intelligence feeds, and target-specific information sources. Output handlers manage result formatting, documentation generation, and report compilation for client delivery.

## Implementation Timeline Overview

The following timeline presents an integrated view of the five development phases, highlighting key milestones and dependencies. This roadmap spans approximately four years from current foundation work through full Manager of Managers deployment, with iterative capability improvements throughout.

| Phase                        | Timeframe       | Key Deliverables                                                   |
|------------------------------|-----------------|--------------------------------------------------------------------|
| Phase 1: Foundation          | Q1-Q2 2026      | MVP production deployment, baseline metrics, operational workflows |
| Phase 2: Benchmarking        | Q2-Q3 2026      | Security-specific benchmarks, model evaluation, selection criteria |
| Phase 3: Model Development   | Q3 2026-Q2 2027 | In-house models, distillation pipeline, training infrastructure    |
| Phase 4: Autonomous Pipeline | Q2-Q4 2027      | 24/7 discovery operations, CVE processing, knowledge systems       |
| Phase 5: Manager of Managers | Q1 2028-Q4 2029 | Swarm teams, human-AI orchestration, scaled operations             |

Table 3: Implementation Timeline Summary

## Success Metrics

Progress throughout the roadmap will be measured against both quantitative and qualitative metrics. Quantitative metrics include the number of unique vulnerabilities discovered, proof-of-concept development success rates, time-to-PoC for disclosed CVEs, and operational cost reduction from local model deployment. Qualitative metrics encompass client satisfaction with deliverable quality, team expertise development, and recognition within the security research community.

Phase-specific metrics provide granular progress indicators. Phase 1 success requires stable production deployment with documented operational procedures. Phase 2 deliverables include benchmark suites with demonstrated differentiation between model capabilities. Phase 3 success requires in-house models achieving comparable or superior performance to teacher models on benchmark tasks. Phase 4 metrics focus on autonomous operation duration and discovery output rates. Phase 5 success is measured by research throughput per human researcher and client satisfaction with managed research outputs.

## Risk Considerations

Several risk factors require ongoing management throughout roadmap execution. Technical risks include challenges in model distillation effectiveness, infrastructure scalability limitations, and integration complexity across framework components. Market risks encompass changes in the threat landscape that alter research priorities, competitive developments that affect differentiation, and regulatory considerations around security research activities. Organizational risks involve talent acquisition and retention, knowledge transfer requirements, and change management as roles evolve toward human-AI collaboration.

Mitigation strategies address these risks through multiple mechanisms. Technical risks are managed through iterative development with frequent validation checkpoints, enabling course correction before significant resource commitment. Market risks are addressed through flexible architecture that accommodates shifting priorities and continuous monitoring of competitive developments. Organizational risks are mitigated through investment in team development, clear career progression paths that evolve with organizational changes, and documentation practices that preserve institutional knowledge.

This roadmap provides a structured path from our current foundation to a world-class autonomous security research capability. Success requires sustained commitment to technical development, organizational evolution, and the balance between automation efficiency and human creativity that distinguishes exceptional research organizations. Regular roadmap reviews will assess progress, adjust timelines based on developments, and incorporate learnings from each phase into subsequent planning.